Navigating the promises and pitfalls of AI as individuals and as a society

Perspectives from building and using AI in everyday life



Charles C. Hall & Lisa Harper, DSc

TVOR · October 2026

bit.ly/4zficKD

About us

Charles C. Hall

- TVOR resident
- Decades of experience building websites, tools, and maps for Hampshire County
- Author of https://historichampshire.org

Lisa D. Harper, DSc

- 30+ years building AI for the U.S. Government
- Adjunct Professor, Shenandoah University
- Founder Ai Craft, Community AI education, consulting & engineering

What we will talk about

> Our focus: how AI affects **you** as an individual, and **us** as a community and society.

1

The worrisome downsides of AI

2

The huge potential and positive impacts of AI

3

How it works

4

Where we're headed



The car didn't wait for the horse to agree. AI is **here to stay**: the worries are real, but **doom is not**. We choose the rules of the road.

Negatives

The Internet's Five Layers of Dependence

Daily life	Phones, streaming, banking apps, health portals
Money	Card payments, bank transfers, interbank settlement
Commerce and logistics	Ordering, inventory, freight tracking, warehouses
Physical infrastructure	Power, water, pipelines, rail, remotely managed plant
Shared foundations	**A few cloud providers and telecom backbones**

Illustrative map for briefing purposes only. Bottom layer supports every layer above it.

EXPOSURE

One provider's failure reached tens of thousands of firms

70,000

organizations disrupted, about, in the AWS outage of 20 October 2025

2,000+

large enterprises among them

8 of 9

major AWS outages since 2012 struck one region, Northern Virginia

Estimated insured losses from the October 2025 outage: $38M to $581M, clustered toward the lower end.

Sources: CyberCube via Risk & Insurance (Oct 2025); OutageCost.com, an independent outage tracker

Six Cyber Incidents with the Greatest Apparent Reach 2024–2026

Incident	Number potentially affected
Chinese data exposure (2025)	Potentially millions to hundreds of millions; 4 billion records reported, not unique people
Ticketmaster / Snowflake breach (2024)	Up to 560 million records claimed; unique individuals not confirmed
Change Healthcare (2024)	192.7 million people reported by UnitedHealth (July 2025)
AT&T call-record breach (2024)	About 110 million customers estimated; exposed individuals not separately confirmed
PowerSchool (disclosed 2025)	About 62 million students + 9.5 million teachers reported by security outlets; not confirmed by company
Canvas / Instructure (2026)	Platform serves about 30 million users; attackers claimed data involving 275 million people, unverified

Source: https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents

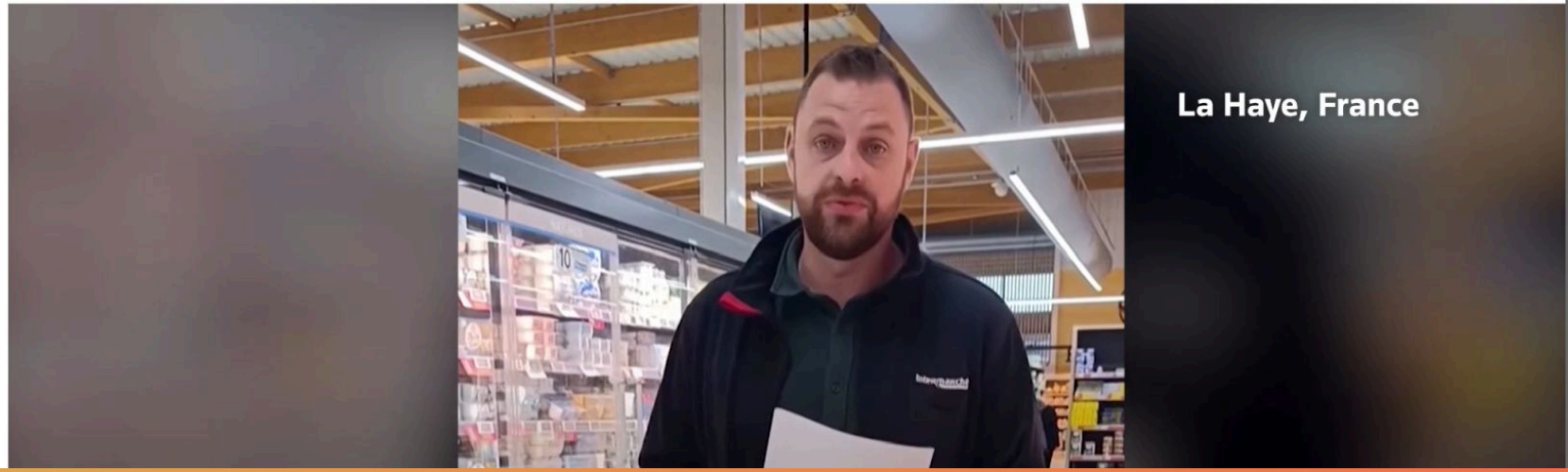
Figures are not directly comparable: records, customers, platform users and unique people are different measures.

https://www.reuters.com/business/retail-consumer/french-supermarket-chain-intermarche-hit-by-cyberattack-150-stories-targeted-2026-10-12/

Hackers heat French supermarket fridges, sparking fires and spoiling food

By **Leo Marchandon** and **Gabriel Stargardter**

October 12, 2026 9:57 AM EDT · Updated October 13, 2026




abcnews.com/Politics/pentagon-breach-exposed-sensitive-data-3
abc NEWS
Live
Video
Shows
ShopGMA
Pentagon breach exposed sensitive data on nearly 3 million people
The breach included Social Security numbers and details about jobs people held.

FBI grapples with fallout from massive data breach

Updated Sept. 29, 2026

UPDATED SEP 29, 2026

By Sean Lyngaas, Evan Perez



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By Biltmore

Positives

How much food could AI save in U.S. grocery retail?

A research-based estimate for AI-assisted ordering and demand planning

907,372 tons
of food waste prevented per year

1.81 billion lb
of food — the same estimate in pounds

Bottom line

If grocery retailers broadly implemented the AI ordering approaches studied in the pilots, ReFED estimates that about 907,000 tons of food waste could be prevented each year.

Sources: ReFED; U.S. Food Waste Pact. See final slide.

Neal K. Shah: In rural North Carolina, the closest hospital may soon be a smartphone

Posted 6:00 AM May 19, 2026

A healthcare tech expert's proposal to give rural North Carolina a tech-driven solution for its lack of emergency care that many communities face.

Shah's argument is that remote technology alone won't work in places with no staff. He writes that Hospital at Home "runs on people," meaning people willing to drive forty miles to take vitals, change a dressing, or sit with a confused elder. His fix is to pair technology with a human workforce drawn from local campuses. He proposes a "Digital Care Corps": a platform that would send a vetted nursing student to a home-hospital patient in a place like Beaufort County, supervised remotely by an ECU clinician, and paid for through a mix of Medicare, Medicaid, and Rural Health Transformation money.

In this model, AI and telehealth become the coordination and supervision layer, and students supply the hands. That's more realistic than "AI nurses," and it ties into real policy.

“Foundation models” trained on huge satellite archives

NASA/IBM Prithvi – first AI model trained specifically on satellite imagery; spots flood boundaries and wildfire burn scars

NASA SatVision-TOA – learned from 100 million NASA satellite images; openly released for anyone to use

Google Satellite Embeddings – summarizes what satellites saw at any place and time (preview, July 2026)

Plain-language search – describe what you’re looking for and get matching satellite images back

Sources

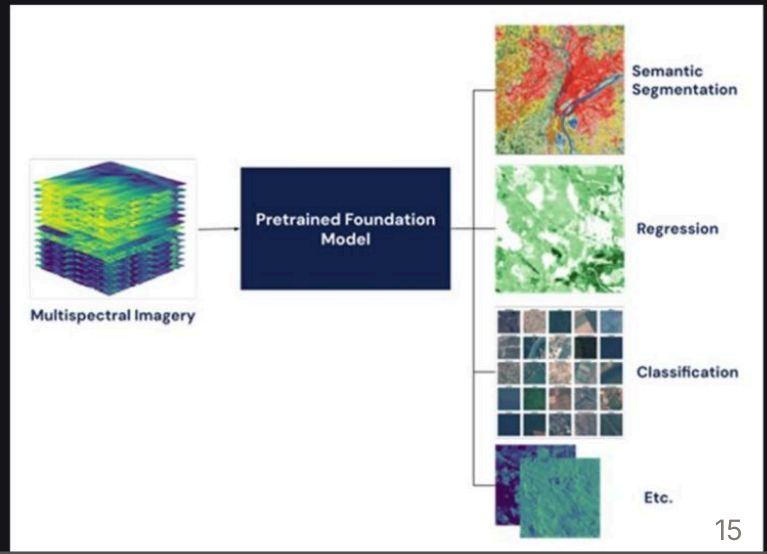
ESA – Nepal flash flood imaged by satellites
Infrastructure Today – Suhora Nepal flood platform
Pléiades Glacier Observatory – 2026 elevation maps
Glacier data source roundup (bullwinkle.js.org)
NASA Earthdata – IBM geospatial foundation model
R&D World – NASA SatVision-TOA
New Space Economy – Earth observation foundation models
Maptive – How AI is changing location intelligence in 2026

3-billion-parameter vision transformer
on a 100-million-image MODIS TOA dataset

OPERATIONAL

SatVision-TOA Geospatial Foundation Model (SatVision-TOA)

SatVision-TOA demonstrates the untapped potential of leveraging moderate- to coarse-resolution data for deep learning in Earth observation. By training a 3-billion-parameter vision transformer on a 100-million-image MODIS TOA dataset, it...
establishes a scalable, open-source foundation for advancing atmospheric science



AI and Satellite Projects

ESA
Nepal flash flood imaged by satellites
esa.int

NASA Earthdata
IBM geospatial foundation model (Prithvi)
earthdata.nasa.gov

Infrastructure Today
Suhora opens free Nepal flood platform
infrastructuretoday.co.in

R&D World
NASA SatVision-TOA trained on 100M images
rdworldonline.com

Pléiades Glacier Observatory
2026 glacier elevation-change maps
legos.omp.eu

New Space Economy
How Earth observation foundation models work
newspaceeconomy.ca

Glacier data roundup
Glacier datasets and satellite sources
bullwinkle.js.org

Maptive
How AI is changing location intelligence in 2026
maptive.com

More on Prithvi and plain-language search

IBM Research – Prithvi 2.0

Earth Genome – Earth Index now open to everyone

Mongabay – A search engine for the planet opens to the public

Indicator – Step-by-step guide to Earth Index

Hugging Face – EarthEmbeddingExplorer tutorial

MICROSOFT SECURITY FIXES, THEN AND NOW

Frontier AI unleashed a flood of fixes

2002–2004 · THE WORM ERA

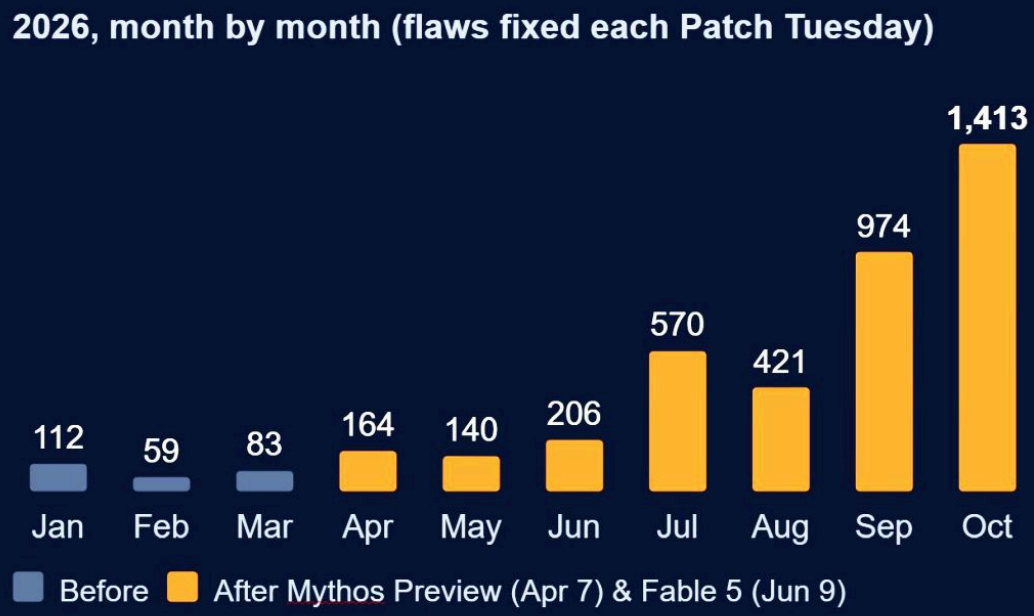
168

security bulletins in 3 years

2026 · JANUARY–OCTOBER

4,100+

flaws fixed in just 10 months



2002–04 counts are bulletins, which often fixed several flaws each; 2026 counts are individual flaws. Sources in notes.

TECHNOLOGY

Hacking attempts rise after flaws found in program affecting 70% of websites

OCTOBER 23, 2026 · 5:37 AM ET

By Kate Bartlett

What has changed in a generation

Then

- Control systems kept apart by air gaps
- Computing run in-house, at many sites
- Outages local and usually short
- Attacks needed scarce specialist skill

Now

- Industrial systems reachable from corporate networks
- Workloads concentrated in a few cloud providers
- One fault can reach thousands of firms at once
- **AI is lowering the skill barrier**

Sources: CSIS on the erosion of air gaps; UK NCSC assessment of AI-enabled intrusion (2025)

What AI changes

More actors

Capability spreads from a few well-resourced states to a wider range of state and non-state actors.

Faster pace

Analysts expect AI to speed the search for weaknesses, shrinking the time defenders have to respond.

Larger surface

AI systems built into infrastructure add new points of failure and new routes for attack.

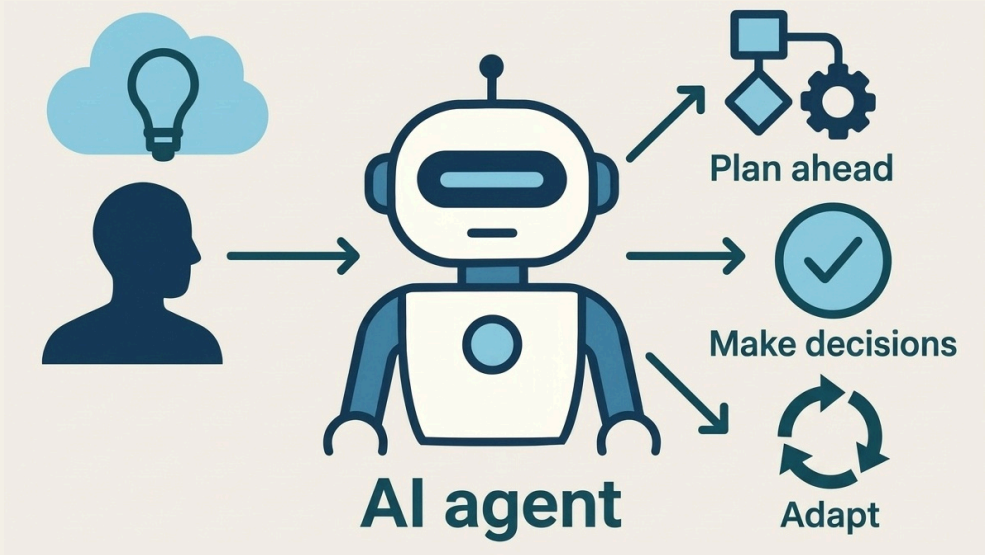
Caution: Public evidence of AI-enabled attacks is still thin, and much commentary comes from vendors and advocacy groups. Weight government and primary reports more heavily.

Sources: UK NCSC forward assessment (2025); commentary on AI acceleration of cyber operations

How does AI work?

What is AI?

- AI is nearly 70 years old!
- Defined as, "the science and engineering of making intelligent machines."
- It's called "**artificial**" intelligence to distinguish it from "**human**" intelligence.
- AI that you interact with using language is *agentic* - it uses tools to remember and perform tasks.
- ***Language is finally the interface for talking with computers!***



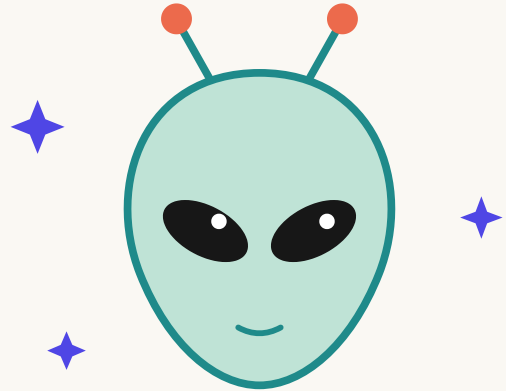
Is it normal technology or something very different?



Normal technology

Like electricity: it changes everything, but slowly, and people stay in charge

or



Something alien

A new kind of mind that could outthink us and act on its own

AI you already use

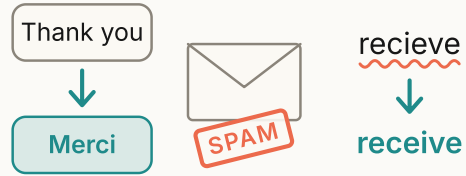
Sees
computer vision



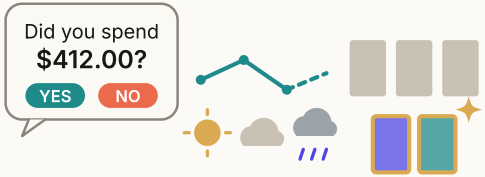
Hears & speaks
speech recognition



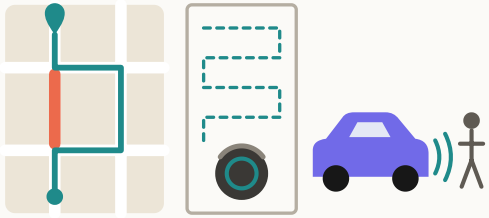
Reads & translates
language processing



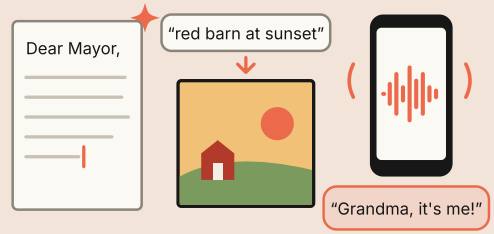
Predicts
machine learning



Plans & moves
planning, robotics

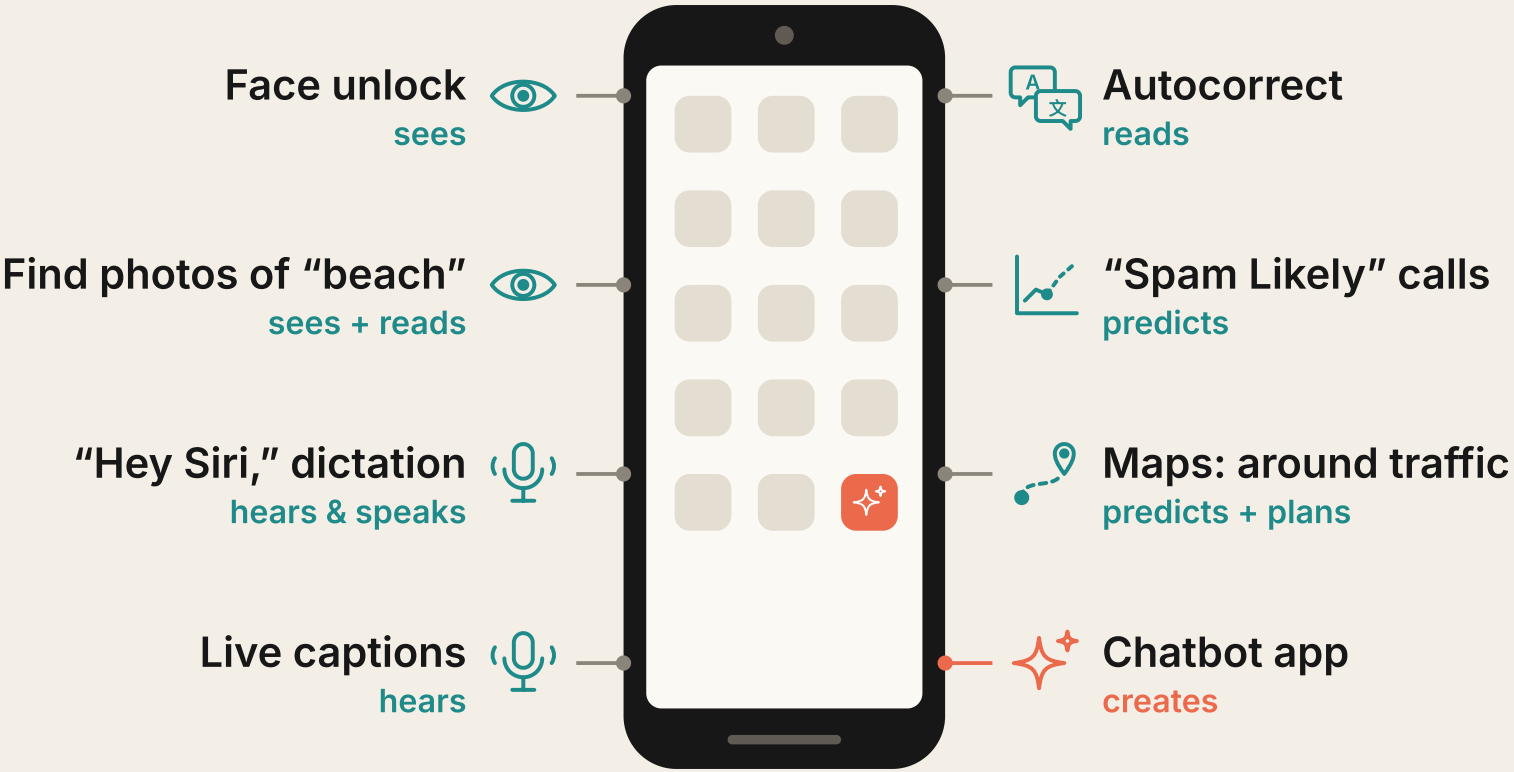


Creates
generative AI



Today most people say "AI" and mean **only the last tile**. It's one of six.

One product, many AIs

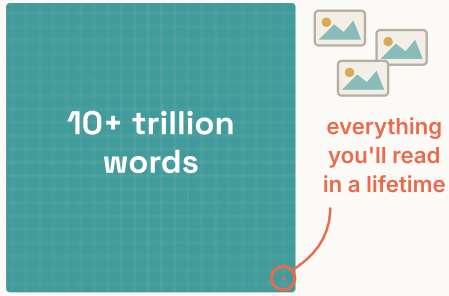


Each feature is a different AI part. The chatbot is one app on the phone.

What is a large language model?

1 Training

reads a vast library of text



≈ 100 million books + billions of images

Reading nonstop, a person would need about 75,000 years.

learns

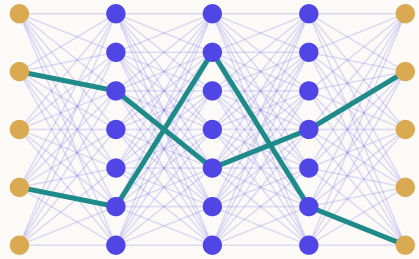
then tuned

people rate it

2 Patterns, frozen

frozen

stored as billions of weights



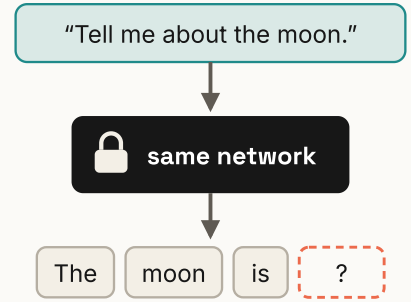
Thick lines: patterns it learned

When training ends, the weights are locked. Your chats don't change them.

you ask

3 Inference

generates text from your words



Predicts the next word, then the next

It uses the patterns. It doesn't look anything up.

Training: once, at enormous cost.

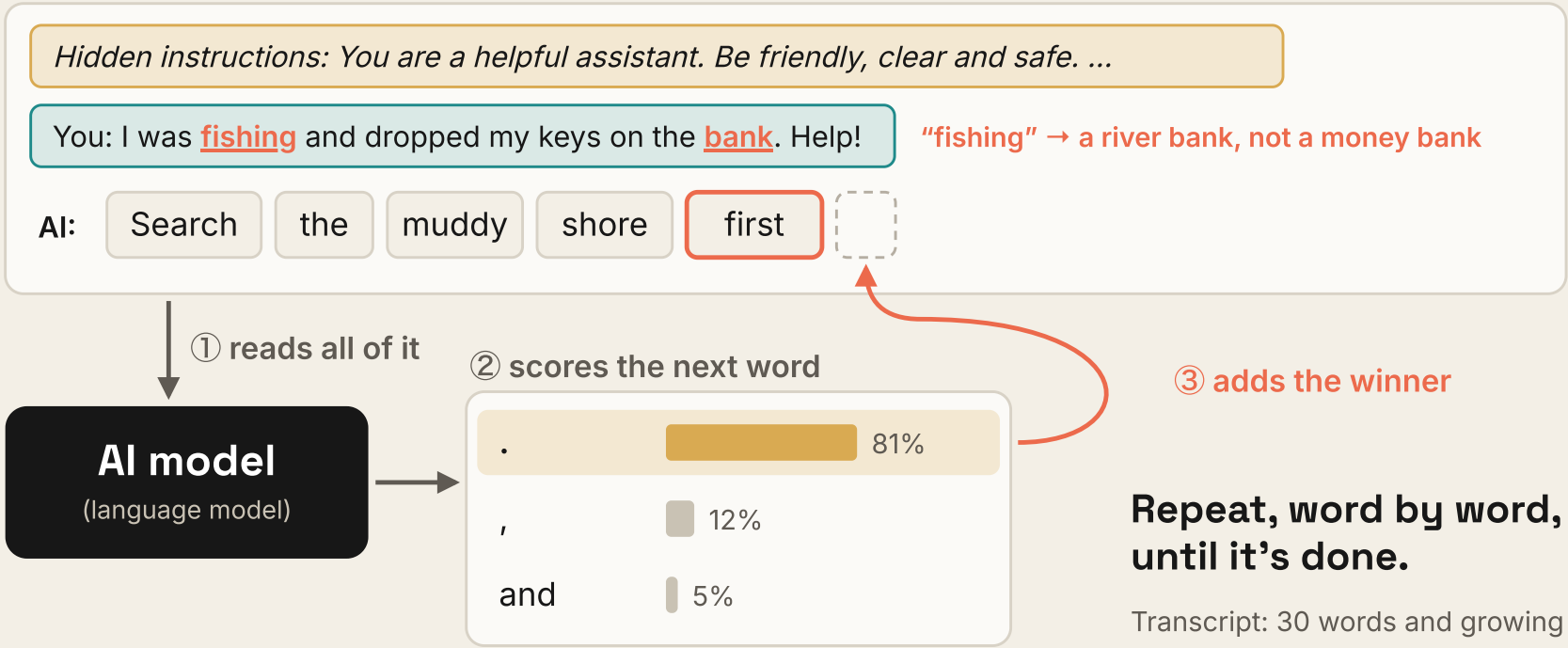
Months on thousands of chips. Then the weights freeze.

Inference: every message, every chat.

The same frozen network, used again and again.

A chatbot writes one word at a time

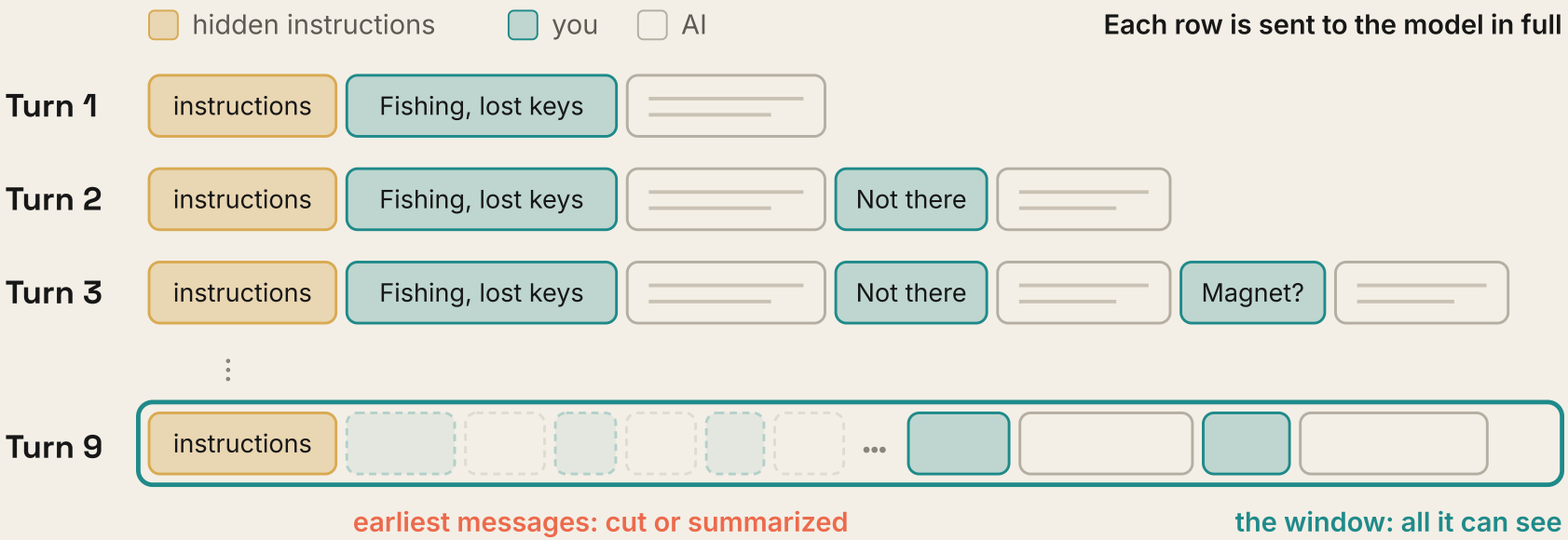
The transcript: everything the model reads



Repeat, word by word, until it's done.

Transcript: 30 words and growing

Every turn, it rereads the whole conversation

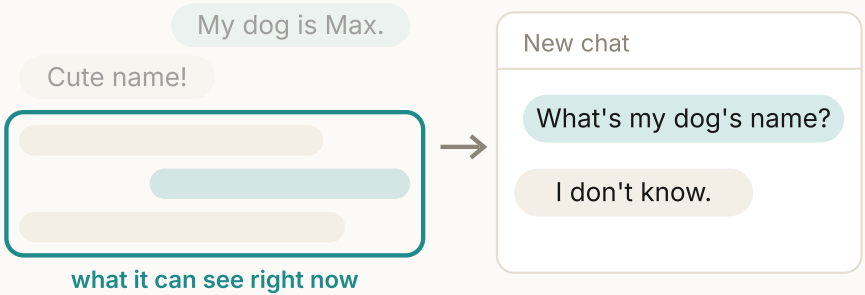


The instructions stay. Your first request may not, and without “fishing”, “bank” is a guess again.

LLM limitations

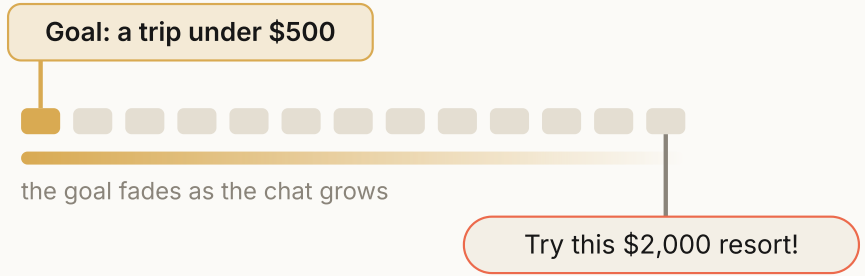
1 No memory

only sees this conversation



2 Loses the thread

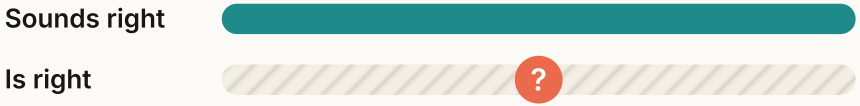
long chats drift from the goal



3 Confidently wrong

built to sound right

Absolutely! Smith & Lee (2019) proved this in the *Journal of Rural Health*, page 42.



fluent is not the same as factual

4 Eager to please

tells you what you want to hear

I think my plan is great!

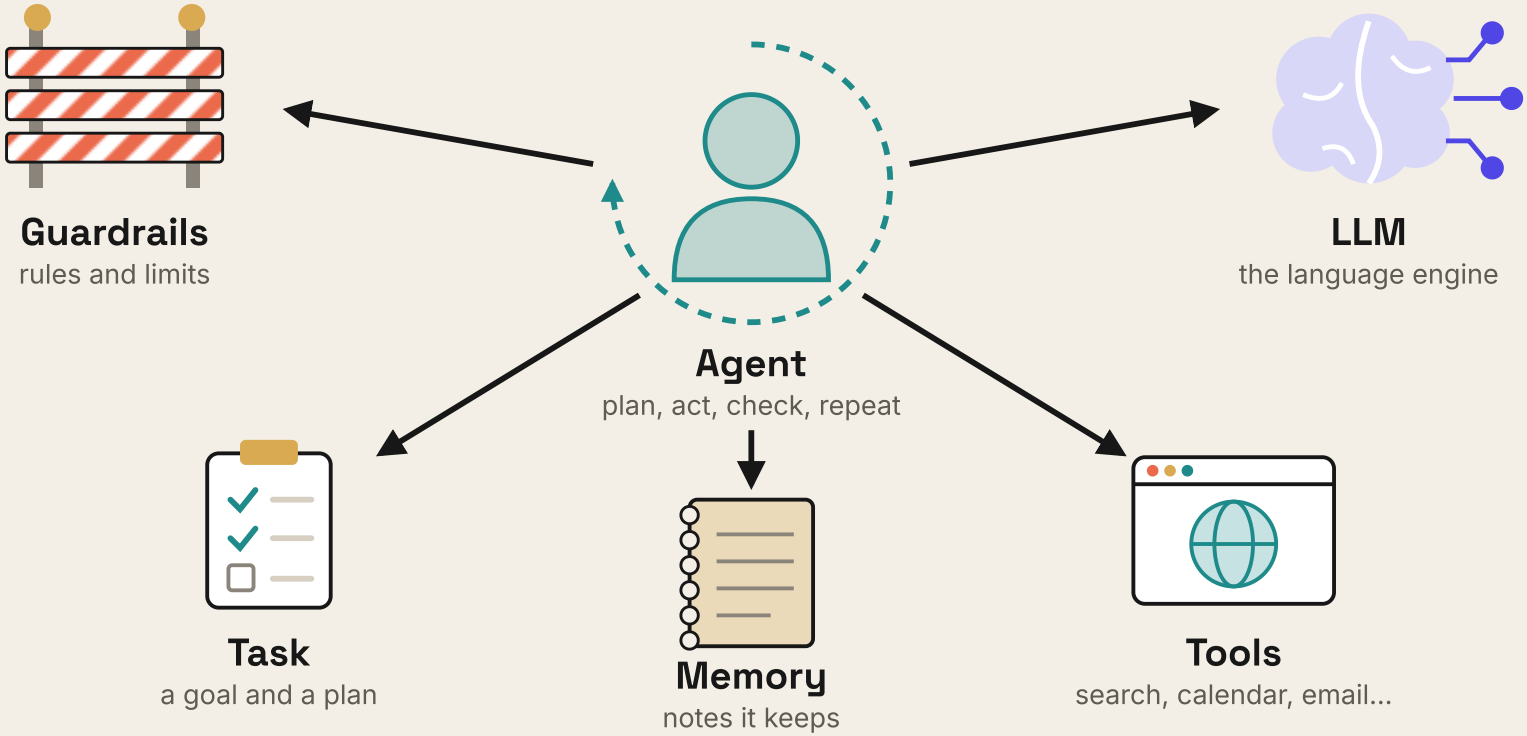
It's brilliant. Go for it!

Hmm, maybe it's not so great.

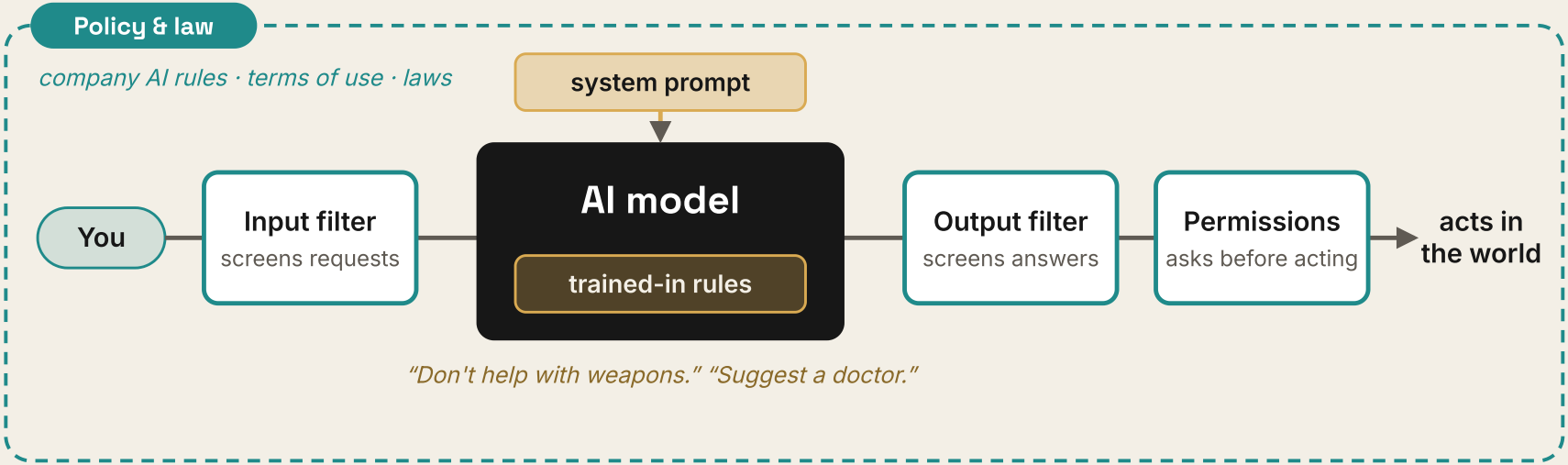
You're right, it has real flaws.

same plan, opposite verdict

Extending LLMs to build agents



How guardrails work

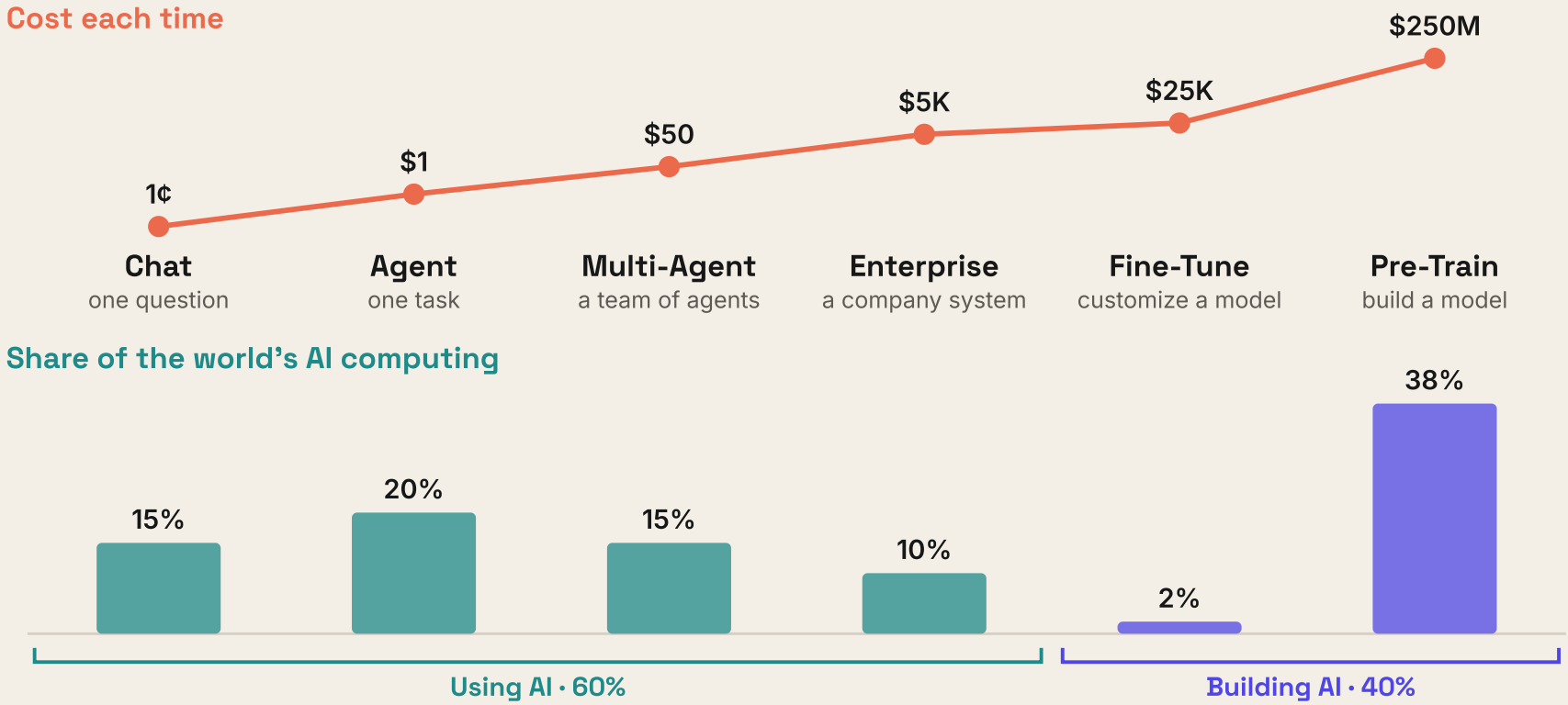


Soft: words and habits inside the model.
Strong, but it can be talked around.

Hard: code and rules outside the model.
Arguing with the AI doesn't move them.

The cost of AI and agents

A chat costs about a penny. Building the model costs hundreds of millions.



Where are we headed?

Getting it right: AI that checks its work

A bat and a ball cost $1.10 together.
The bat costs $1.00 more than the ball.
How much is the ball?

System 1
fast · automatic · fluent

10¢

sounds right, isn't

Today's chatbots, by default

System 2
slow · careful · checks

ball + (ball + $1.00) = $1.10
2 × ball = $0.10
ball = 5¢ ✓

What's being added

ADDING SYSTEM 2



Calculate, don't guess
language + real logic and math



Brain-like parts
memory, planning, what-ifs



The physical world
not just text

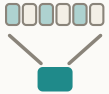
Yang: fewer confident wrong answers.

Yin: reliable AI tempts us to stop checking.

Efficiency & energy: the power behind AI

1 Leaner models

Today: rereads the whole conversation for every word



Smarter with long chats
less rereading, less forgetting



On your own devices
smaller models, no data center

2 Less power per answer

Today: most AI computing goes to answering, every day



Chips built for answering
not all-purpose graphics chips



Cooling
packed racks run hot; water is local



Power
new lines, plants, even small reactors

Yang: each answer gets cheaper and cleaner.

Yin: cheaper means more use. Where data centers go is decided locally.

Conclusion

Worth remembering

WHAT YOU CAN DO



Keep chats short

New topic, new chat. Less energy, better answers.



Know how it works

Enough to choose your tools wisely.



Trust, then verify

AI will get more reliable. Checking stays your job.

WHAT WE CAN DO TOGETHER



Pause before agents

Ask what work, school and town let them do.



Energy & infrastructure

Power, land and water are decided locally.

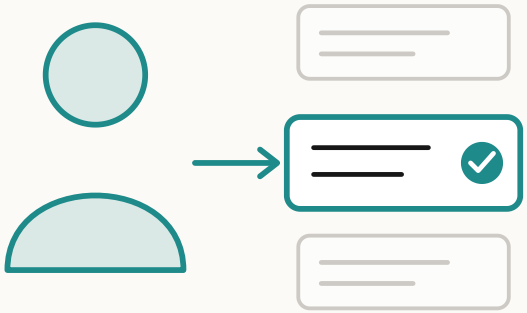


Weigh yin and yang

Who benefits, and what does it cost?

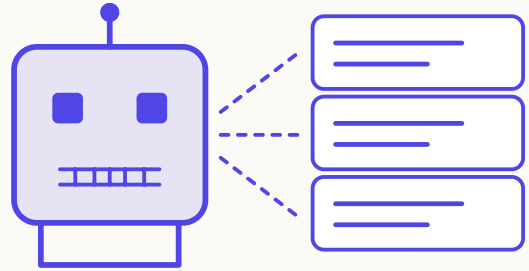
AI is a tool. The decisions stay ours.

Humans decide



Weigh the options. Make the call. Own the result.

AI assists



Drafts, options and warnings, faster than we can. Never the final say.

But people build AI, too. Some are racing to take the human out of the loop. Those choices are made in companies, not here. Keep asking: **who decided, and who checks?**

References

Framing: what is AI?

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Cost and compute

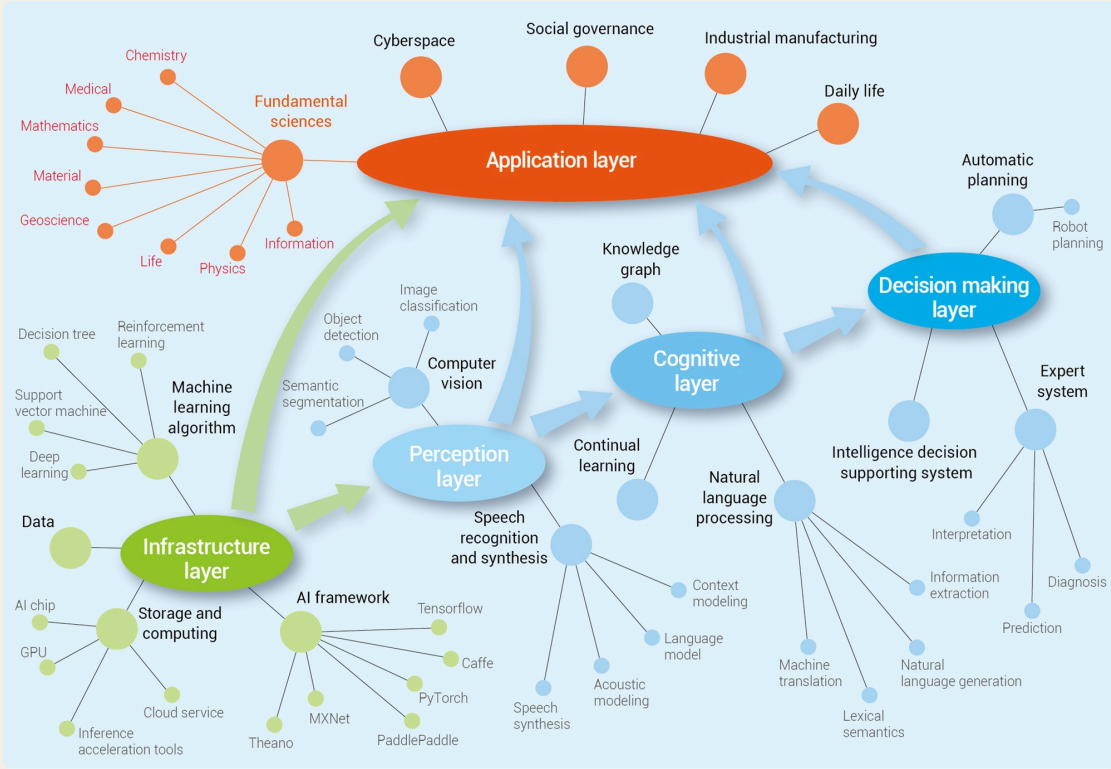
- Hall, C. Navigating AI: cost and compute sources. http://www.cchall.org/temp/navigating-links.pdf

Practical tools

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Backup

AI is not a monolith - it's comprised of many evolving parts



Xu et al. (2021), "Artificial intelligence: A powerful paradigm for scientific research," *The Innovation* 2(4):100179. CC BY-NC-ND 4.0

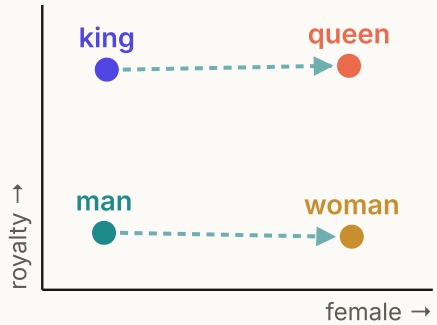
Revolutionary ideas for language interfaces (Large Language Models)

1 Meaning as numbers

word embeddings

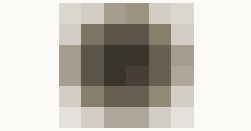
	royalty	female	...
king	0.93	0.06	...
queen	0.95	0.94	...
man	0.08	0.05	...
woman	0.06	0.95	...

real models: hundreds of unlabeled columns

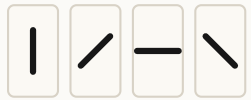


2 Layers find structure

deep networks



Layer 1
light & dark



Layer 2
edges



Layer 3
parts



Layer 4
objects

"cat"

3 Context shapes meaning

transformers

We fished from the **river bank**.

shore

I took **cash** to the **bank**, money

but **it** was closed. it = bank

same word "bank", different meaning